Pancreatic cancer is usually diagnosed late, but the metabolic dysfunction associated with it does not start at diagnosis. Changes in glucose regulation and body composition can appear months or years earlier, scattered across registry records, cohort data, and imaging that are rarely analysed together. Treating that scattered signal as one multimodal risk problem across time is a difficult and interesting problem to work on.

I have recently completed my Master's degree in Data Science at Uppsala University, where I gradually focused my interests on healthcare and machine learning. What initially drew me to the field was how data-driven systems can improve healthcare while remaining trustworthy and useful in practice. Early on, I wrote about bias in medical AI systems and how existing limitations in healthcare research risk being amplified as increasingly capable models are deployed. Since then, I have become increasingly interested in how data quality, representation, and modelling choices influence the conclusions we draw from biomedical data.

Through a health innovation course, I studied the European health technology ecosystem and the challenges involved in bringing new technologies into clinical use. In another project, I investigated healthcare data interoperability challenges in Sweden, examining how information moves between systems and organisations. More recently, I started a pilot collaboration with the Haematology and Oncology Department at King Faisal Specialist Hospital in Saudi Arabia, exploring AI-assisted interpretation of electrophoresis results. While these experiences differ in scope and technical depth, they all reinforced the same observation: healthcare data is rich, fragmented, and often provides only a partial view of the underlying biological processes.

My strongest technical experience comes from machine learning and software development. My master's thesis focuses on multimodal machine learning for harmful content detection, combining text, image, and contextual information within a unified modelling framework. The work required developing custom PyTorch architectures, designing representation learning approaches, handling large datasets, and making modelling decisions under significant ambiguity. Different domain, but key skills are transferable: integrating heterogeneous data sources, learning useful representations from incomplete information, and identifying which information is genuinely predictive versus merely correlated.

What this position asks for, combining a population-based cohort, a national registry-based cohort, and a clinical imaging cohort into one risk model, overlaps with the kind of problem I already spend most of my time on. Before any model can be fit, the data has to be understood: different collection protocols, time resolutions, and missingness patterns across each source, and how that affects which signals can be trusted. My thesis involved the same kind of work in a different domain, figuring out what a multimodal dataset could actually support before building anything on top of it.

My academic background is in data science. I do not have formal training in epidemiology, biostatistics, or cancer epidemiology specifically. What I bring instead is comfort with large, heterogeneous datasets, a structured and self-directed way of working, and a specific interest in the statistical and machine learning methods this position centres on. I expect to build the

cancer-specific epidemiological and clinical knowledge through the programme and the group, the same way I have picked up every domain I have worked in so far. I am available to start from September 2026.